

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 04 · LECTURE NOTES

The Price of Anarchy

What selfish behaviour costs, measured.

MODULE OUTCOMES

Define the price of anarchy and the price of stability.

Compute both for small examples.

Explain the smoothness framework.

State the routing bound and its conditions.

Use the measure to decide whether to intervene.

1 The definition

price of anarchy = (cost of worst equilibrium) / (optimal cost); price of stability = (cost of best equilibrium) / (optimal cost)

- **The price of anarchy is a worst case over equilibria, so it is a guarantee:** however badly the agents coordinate among the stable outcomes, the result is no worse than that factor.
- **The price of stability is a best case, so it is an aspiration:** it is what you could achieve if you were able to suggest which equilibrium to play — **relevant whenever a designer can propose a starting configuration** and the agents have no reason to move away from it.
- **Both are at least 1, and both depend on the solution concept used** — a price of anarchy taken over **mixed equilibria may exceed the one over pure equilibria**, simply because there are more mixed equilibria to be worst over (Module 02 §2's hierarchy).
- **Which ratio you want depends on whether you can coordinate the agents at all** — and in most engineered systems you can suggest a default configuration, which makes the price of stability the more relevant number.

AND IT TURNS A QUALITATIVE COMPLAINT INTO A NUMBER, WHICH IS WHY THE MEASURE MATTERS

'Selfish behaviour is inefficient' was known for decades and was not actionable — 'selfish routing costs at most 33% more than optimal when latencies are linear' is a bound you can design against, budget against, and check.

So the measure converts a general worry into an engineering quantity — which is this semester's and this program's recurring move (CSCE 679 Module 02's measured channel ranking, CSCE 632 Module 13's substituted measurement).

And a small price of anarchy is a licence to do nothing: if the equilibrium is within a few percent of optimal, centralised control is not worth its cost in latency, complexity, and the new incentive problem it creates (§4).

Which is the genuinely practical use of the concept — it tells you when to intervene and when the decentralised system you already have is good enough, which is a question engineers face constantly and usually answer by intuition.

2 Examples

PIGOU'S EXAMPLE

two parallel roads, one unit of traffic
 road 1: latency 1, regardless of load
 road 2: latency = x , the fraction using it
 equilibrium: everyone takes road 2 (it is always
 at most 1), total cost 1
 optimum: split half and half, total cost $3/4$
 price of anarchy = $4/3$

AND WITH NONLINEAR LATENCY

replace x with x^d and the ratio grows without
 bound as d grows. The bound DEPENDS ON THE CLASS
 OF LATENCY FUNCTIONS ALLOWED.

LOAD BALANCING, identical machines

the price of anarchy is small and constant

AND UNBOUNDED CASES EXIST

some network design and scheduling games have
 equilibria arbitrarily worse than optimal, which
 is precisely the case for intervening

The bound depends on the class of latency functions allowed — which is the condition most often dropped when the 33% figure is quoted, and it is not a technicality: with polynomial latencies of degree d the ratio grows roughly like $d / \ln d$, which is unbounded. Pigou's example is worth working by hand: it is two roads and four lines of arithmetic, and it contains the whole phenomenon.

3 Smoothness

A SMOOTHNESS ARGUMENT BOUNDS THE PRICE OF ANARCHY WITHOUT ANALYSING THE EQUILIBRIUM AT ALL

If the game satisfies a particular inequality relating any outcome to the optimal one — roughly, that each agent deviating toward its role in the optimum does not cost the system too much — then a bound on the price of anarchy follows mechanically from the definition of equilibrium.

And the resulting bound applies to every solution concept that satisfies the deviation condition — pure Nash, mixed Nash, correlated equilibria, and the time-averaged outcomes of no-regret learning (Module 12 §3) — because all of them satisfy the one inequality the proof uses.

Which is the remarkable part: one inequality, proved once about the game itself, bounds an entire family of behaviours — including ones you never analysed and ones that had not been defined when the bound was proved.

And it means the bound survives agents who do not play equilibrium at all — which answers Module 03's objection directly: you no longer need the agents to solve a PPAD-complete problem for the guarantee to hold, only to not persistently regret their choices. That is why the framework mattered rather than being one more bounding technique.

4 Using it

- What is the benchmark? — 'optimal' means optimal with respect to a stated objective, and the objective is a design choice: total cost, maximum cost (makespan), or some fairness criterion give different optima and different ratios.
- Which equilibrium concept? — and does the bound hold for the behaviour you actually expect from your agents? §3's answer is frequently yes, which is the smoothness framework's practical payoff.
- What class of cost functions? — which is where the bound comes from (§2), and is the condition that gets dropped in summary, turning a conditional theorem into a false general claim.
- Is the worst case the relevant case? — a bad worst-case bound with good typical behaviour may not justify intervention, which is what Project 1 measures by reporting the distribution of ratios rather than the maximum.
- And what would the centralised alternative actually cost? — in latency, in trust, in operational complexity, and in the incentive to misreport to the coordinator (Module 01 §2). The centralised alternative has its own incentive problem: agents now report to the coordinator, and the coordinator's input is strategic too — which is the trap in 'just centralise it'.

AND THE HONEST SUMMARY

The price of anarchy is frequently small, which is genuinely good news — decentralised systems are often nearly optimal — and that is worth knowing before building a coordinator that will be expensive and will introduce a new strategic surface.

And where it is large, the measure tells you how much a fix is worth — which bounds the sensible budget for the intervention and makes the decision quantitative rather than architectural taste.

Plus the bounds are worst case over instances — so a measured distribution over your actual instances is more useful than the theorem for an engineering decision (Project 1's requirement), while the theorem is what tells you the measurement cannot be arbitrarily bad.

Which is this module's version of the program's rule: name the benchmark, the solution concept, and the function class — and a ratio quoted without all three is not a number, which is the same shape as CSCE 640 Module 13's three requirements.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Koutsoupias & Papadimitriou — Worst-case equilibria (free)	§1 — where the measure was introduced, in a load balancing setting.
Roughgarden & Tardos — How bad is selfish routing? (free)	§2's routing bound — with the latency function condition made explicit, as it should be.
Roughgarden — Intrinsic robustness of the price of anarchy (free)	§3's smoothness framework — and the extension to no-regret outcomes is the result that matters.
Nisan et al., chapters 17 to 19 (free PDF)	§§2 and 4 — the bounds for routing, load balancing, and network design.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Define both ratios** and give a game where they differ.
- 2 **Work Pigou's example** by hand, both costs.
- 3 **Replace the latency with x^3** and recompute the ratio.
- 4 **Find the ratio for x^d** as a function of d .
- 5 **Compute the price of anarchy** for a two-machine load balancing game.

- 6 Find a game with unbounded price of anarchy.
- 7 State the smoothness inequality and what it gives you.
- 8 Explain why the bound extends to no-regret play.
- 9 For one real system you know, name the benchmark, the concept, and the function class.
- 10 Estimate what centralising it would cost, including the new incentive problem.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Define both ratios and say which is a guarantee.
- 2 Why does the concept used change the number?
- 3 What did the measure contribute over 'selfishness is inefficient'?
- 4 When is a small price of anarchy actionable?
- 5 Work Pigou's example and give the ratio.
- 6 What happens with nonlinear latencies?
- 7 Which condition is usually dropped when quoting the bound?
- 8 What does a smoothness argument prove, and over what family?
- 9 Why does that answer the computability objection?
- 10 Give the five questions for applying this to a real system.